

## GLOSSARY

**Acceleration** is the time rate of change of velocity and is a vector quantity.

**Atmospheric Pressure** or barometric pressure is the force exerted by the air (its weight) on unit area.

**Bar magnet and solenoid** have same pattern of magnetic fields.

**Barometer** is a device used to measure atmospheric pressure.

**Centre of gravity** is a single point where the whole weight of an object appears to act.

**Centre of mass** of the body is the point about which mass is equally distributed in all directions.

**Centripetal force** is the force that compels a body to travel a circular path. It may be electric, gravitational, or any other force.

**Density** is mass per unit volume

**Diamagnetic materials** have zero net magnetic fields per atom.

**Displacement** is the shortest distance from the initial and final position of a body.

**Distance** is how much ground an object has covered during its motion.

**Drag Force** is the force applies acting opposite towards the motion of the object which is moving submerge in a certain fluid.

**Dynamics** is the branch of mechanics in which we discuss the motion of bodies along with causes of motion of bodies.

**Earth** has a magnetic field which resembles with the field of a bar magnet and shield all the life on it from cosmic rays and radiation from the Sun and outer space.

**Earth's geographical and magnetic poles** are not same but inclined at an angle of  $11.3^\circ$ .

**Efficiency** is the ratio of useful energy or work output to the total energy or work input.

**Elasticity** is the property of a body, which enables the body to regain its original dimension when the deforming force acting on the body is removed.

**Energy** is the capacity of a body to do work.

**Engineering** is the process of studying and developing technology.

**Error** is the difference between the measured and actual values.

**Ferromagnetic materials** have large net magnetic field per atom.

**Force** is a physical quantity which moves or tends to move a body, stops or tends to stop a moving body or which tends to change the speed and direction of a moving body.

**Frictional force** is the force that resists motion of objects on a surface.

**Graph** is a line straight or curved that shows relation between two quantities out of which one varies as a result of change in other.

**Gravitational Field Strength** is defined as the force per unit mass that earth exerts on a body.

**Gravitational potential energy** is the energy possessed or acquired by an object due to a change in its position when it is present in a gravitational field and is given by  $E = mgh$

**Heat** is the form of energy which is transferred from one body to another body due to the difference in temperature.

**Hooke's law** states that within elastic limits the extension (or compression) is directly proportional to the force applied.

**Hypothesis** is a tentative assumption or explanation made before any research that can be verified by further investigation.

**Impulse** is a certain amount of force you apply for a certain amount of time to cause a change in momentum

**Induced magnetism** is the phenomenon in which a material is magnetized by placing it inside an external magnetic field.

**Inertia** is a property of matter by which it remains at the state of rest or in uniform motion

**Kinetic Energy** is the energy of an object due to its motion and is given by  $E = \frac{1}{2}mv^2$

**Law** is a statement that summarizes an observed regularity or pattern in nature and gives the relationship between variables.

**Law of conservation of energy** states that energy can neither be created nor destroyed. It can be converted from one form to another or transferred from one body to another, but the total amount remains constant.

**Law of Conservation of Momentum** states that if there is no external force applied to a system, the momentum of that system remain constant.

**Like poles of magnets** always repel each other, while unlike poles attract each other.

**Liquid Pressure** is the force exerts by water on the walls of the container from inside

**Magnetic domain** is the group of atoms whose N-poles are aligned in the same direction.

**Magnetic field** is the region of space around a magnet in which it is attracted by the magnetic pole of opposite polarity.

**Magnetic field lines** originate from the north-pole and terminate at south-pole.

**Magnetic materials** are always attracted by magnets and other magnetic materials.

**Magnetic shielding** is the phenomenon in which some region of space is made free from magnetic field.

**Magnets** are always found in dipole, i.e. they have an N-pole and an S-pole.

**Mass** is the quantity of matter in a body.

**Migrating birds** and animals use Earth's magnetic field to navigate.

**Moment of a force or Torque** is the measure of an object tendency to rotate about some point O.  
*Moment of a force = force  $\times$  perpendicular distance of the force to the point.*

**Momentum** is the product of mass and velocity. It is a vector quantity.

**Monometer** is one of the most accurate devices for measuring pressure, including atmospheric pressure in the lower ranges.

**Newton's First Law of Motion** states that everybody continues in its state of rest or uniform

motion in a straight line unless an external net force acts upon it.

**Newton's Second Law of Motion** states that whenever a net force acts on a body, it produces acceleration in the direction of the net force. The acceleration is directly proportional to the net force and inversely proportional to the mass of the body.

**Newton's Third Law of Motion** states that to every action there is an equal and opposite reaction.

**Orbital velocity** is the speed of an object revolving around another heavy object in an orbit.

**Paramagnetic materials** have small net magnetic field per atom.

**Pascal's Principle** states that if the pressure at one point of a confined fluid is increased by an amount, the pressure increases by the same amount at all other parts throughout the fluid.

**Permanent magnets** are those magnets which behave like a magnet even in the absence of current.

**Physical quantities** are measurable quantities

**Physics** is the branch of science which deals with the study of matter and energy.

**Plasma** is called fourth state of matter which is the ionized state of matter.

**Position** is the distance and direction of a body from a fixed reference point.

**Potential Energy** is the energy of an object due to its position.

**Power** is the rate of doing work or rate of conversion of energy.  $P = W/t$

**Prefix** is a word placed before quantities to modify their meaning.

**Pressure** is force applied per unit area.

**Principle of moments** states that for an object in equilibrium, the sum of the clockwise moments taken about the pivot must be equal to the sum of the anti-clockwise moments taken about the same pivot.

**Renewable & Non-renewable Energy Resources** are such that renewable resources can naturally replenish themselves while non-renewable

resources cannot.

**Scalars** are physical quantities with magnitude only.

**Science** is natural explanations for apparent phenomena.

**Scientific Notation** is an internationally accepted way of writing numbers in which

**Screw Gauge** is a device used to measure a fraction of smallest scale division by

**Seven Base SI Units** are metre (length), kilogram (mass), second (time), ampere (current), candela (luminous intensity), Kelvin (temperature) and mole (amount of substance).

**Significant Figures** are the accurately known digits and first doubtful full digit in any sliding another scale over it.

**Speed** is time rate of change of distance and is a scalar quantity.

**Stability** of an object refers to the ability of the object to return to its original position when the force that changed its orientation is removed.

**States of matter** are four, which are named as Solids, Liquids, Gases and Plasma.

**Stop Watch** is an instrument used for measurement of time interval

**System International (SI)** is the system of units which consists of seven base units and a number of

derived units.

**Technology** is tools, including methods, that assist people in accomplishing tasks.

**Temperature** is the measure of degree of hotness or coldness of a body.

**Temporary magnets** are those magnets which behave like a magnet only in the presence of current.

**Terminal velocity** is the maximum constant velocity that a body can achieve while passing through a resistive (viscous) medium.

**Theory** explains how nature behaves under specific conditions.

**Thermometer** is a device which is used to measure temperature.

**Thermometry** is the branch of physics, which deals with the measurements of temperature.

**Vectors** are physical quantities with magnitude as well as direction.

**Velocity** is the time rate of change of displacement and is a vector quantity.

**Vernier calipers** is a device used to measure a fraction of smallest scale division by

**Weight** is the downward force with which the earth pulls a body towards its center.

**Work** is force multiplied by distance moved in the direction of the force  $W = FS$

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## BIBLIOGRAPHY

- Ball, L., Devies, B., Lamb, H., Johnson, P., Morgon, B., Snedden, R., . . . Woolley, S. (2021). Super Simple Physics. New York: DK Publishing.
- Brian, H., Nowikow, I., T. Howes, C., Mantha, J., P. Smith, B., & Bemmell, H. M. (2002). Physics: Concepts and Connections . Toronto: Irwin Publishing Ltd.
- D. Cutnell, J., & W. Johnson, K. (2012). PHYSICS. Hoboken: John Wiley & Sons, Inc.
- Determining the Net Force. (2022, February 11). Retrieved from The Physics Classroom: <https://www.physicsclassroom.com/class/newtlaws/Lesson-2/Determining-the-Net-Force>
- Freedman, R. A., & Young, H. D. (2020). University Physics with Modern Physics . Harlow: Pearson Education Limited.
- Giancoli, D. C. (2014). Physics for Scientists & Engineers with Modern Physics. Essex: Pearson Education Limited.
- Hawkes, R., Iqbal, J., Mansour, F., Milner-Bolotin, M., & Williams, P. (2019, 2014). Physics for Scientists and Engineers: An Interactive Approach, Second Edition. Toronto: by Nelson Education Ltd.
- <https://www.nasa.gov>. (2022, March 19). Retrieved from NASA: <https://www.nasa.gov>
- International Bureau of Weights and Measures. (2022, March 18). Retrieved from International Bureau of Weights and Measures: <https://www.bipm.org/en/home>
- Moment Equations Formulas Calculator. (2022, March 17). Retrieved from AJ Design: [https://www.ajdesigner.com/phpmoment/moment\\_equation.php](https://www.ajdesigner.com/phpmoment/moment_equation.php)
- Serway, R. A., & Faughn, J. S. (2006). Physics. New York: Holt, Rinehart and Winston.
- Speed Distance Time Calculator. (2022, March 19). Retrieved from Calculator Soup: <https://www.calculatorsoup.com/calculators/math/speed-distance-time-calculator.php>
- Urone, P. P., & Hinrichs, R. (2020). Physics - High School. Texas: Texas Education Agency (TEA).
- Walker, J., Halliday, D., & Resnick, R. (2014, 2011, 2008, 2005). Fundamentals of physics. Hoboken: John Wiley & Sons, Inc.
- World Energy & Climate Statistics - Yearbook 2021. (2021, March 12). Retrieved from Enerdata: <https://yearbook.enerdata.net/total-energy/world-consumption-statistics.html>
- Zemansky, M. W., & Dittman, R. H. (1997, 1981, 1968, 1957, 1951, 1943, 1937). Heat and Thermodynamics - An intermediate textbook. New York: McGraw Hill Companies.

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